

A cubic-size barrier for the populated-cluster lifting method

Short asymptotic observation, proposed by the root agent

30 September 2026

Proposition 1. *Suppose two probability measures μ_A, μ_B have disjoint ordered clusters of support, each cluster belonging to only one measure. Suppose each measure integrates polynomials of degree at most m exactly as the uniform probability measure on $[0, 1]$. Then there are at least $m + 1$ changes of color between consecutive nonempty support clusters. In particular, each measure occupies at least $\lfloor (m + 2)/2 \rfloor$ clusters.*

Proof. The signed measure $\sigma = \mu_A - \mu_B$ annihilates every polynomial of degree at most m . If its ordered nonempty support clusters had $r \leq m$ color changes, choose one point in each gap where a change occurs. The product of the corresponding r linear factors, multiplied by a suitable global sign, has the sign of σ on every support cluster. Its integral against σ is strictly positive, a contradiction. There are therefore at least $m + 2$ alternating color blocks, requiring at least the stated number of clusters of each color. $\square$

In a fully fair hybrid with these two variables and m continuous uniform variables, the polynomial exactness hypothesis follows simply by restricting to either specified variable and the m uniforms. The method in `clustered_multi_exceptional_lifting.tex` requires at least m^2 distinct nodes in each cluster. Consequently that unmodified method necessarily uses at least

$$m^2 \left\lfloor \frac{m + 2}{2} \right\rfloor = \Omega(m^3)$$

faces on each of the two specified dice. This is a restriction on that particular lifting criterion. It does not give a cubic universal lower bound and does not exclude two dice of order m^2 by another method.